

BSE4104 EMERGING TRENDS IN SOFTWARE ENGINEERING
GROUP A CAPSTONE

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Controlled Corpus & Provenance Source Register

System Use Case: University Student-Support Case Agent

Baseline Model: Gemini 2.0 Flash (Gemini-2.0-flash), accessed through the Google AI API.

Target Workflow: Grounded Q&A, Timetable/Case Lookup, Draft Case Creation

Engineering Paradigm: L LM Suggests, Deterministic Software Enforces

Mandatory AI Safety & Governance Boundary: In strict adherence to the Project Charter and Week 2 Prompt Engineering Artifacts, the AI agent operates solely within a draft and query boundary. Under no circumstances may the generative model or RAG workflow approve, reject, finalize, or alter a support case status beyond "Pending". All high-impact state transitions, grade updates, and official record modifications are strictly reserved for human departmental staff via deterministic validation pipelines.

Section 1: Formal Provenance Source Register

The table below details the formal provenance source register for the controlled corpus. It enforces a strict

Separation of Concerns between unstructured documents processed by the Retrieval-Augmented Generation (RAG) pipeline and structured JSON datasets consumed exclusively by deterministic code paths.

Doc ID	Title / File	Domain	Source Provenance	Linguistic Register / Formality	System Role
SRC-001	Makerere academic regulation (Undergraduate Retake and Resit Policy)	Academic regulations - assessment	Synthetic, team-authored/recreated, modeled on publicly summarized Makerere academic regulation categories; not copied from or verified against a restricted or official handbook	Formal, regulatory (defines conditions, eligibility, numbered clauses)	Unstructured RAG corpus - grounds retake-policy Q&A (US-1)

SRC-002	Missing Marks and Incomplete Grade Procedure	Academic regulations - records	Synthetic, team-authored, modeled on generic university missing-marks workflows	Formal, procedural	Unstructured RAG corpus - grounds missing marks Q&A; informs case-category suggestion, never case decisions
SRC-003	Course Registration and Add/Drop Policy	Academic regulations - registration	Synthetic, team-authored	Formal, regulatory	Unstructured RAG corpus - grounds registration-issue Q&A (US-1, US-4)
SRC-004	Student Support Case Categories and Escalation Routes	Administrative process	Synthetic, team-authored; category list aligned with departmental case-intake practice; escalation boundary cross-references POL-010	Formal, administrative	Unstructured RAG corpus - grounds category suggestion (US-4) and boundary/redirect messages (US-5)
SRC-005	Grading Appeals and Review Window	Academic regulations - assessment	Synthetic, team-authored	Formal, regulatory	Unstructured RAG corpus - grounds appeals related Q&A; explicitly out of AI decision scope
SRC-006	Disciplinary Referral Process (Summary)	Administrative - disciplinary	Synthetic, team-authored, deliberately high-level	Formal, administrative	Unstructured RAG corpus - used only to trigger boundary/redirect responses (US-5); never for grounded answers on outcomes

SRC-007	Fees, Registration Blocks and Financial Clearance (Summary)	Administrative - finance	Synthetic, team-authored	Formal, administrative	Unstructured RAG corpus - used only to trigger boundary/redirect responses (US-5); fee decisions are out of scope
SRC-008	Timetable Change and Clash Reporting Procedure	Academic administration	Synthetic, team-authored	Formal, procedural	Unstructured RAG corpus - supplements timetable lookup (US-2) with process context
SRC-009	Data Handling and Student Privacy Note	Governance / privacy	Synthetic, team-authored; consistent with Project Charter 7 Constraints and US-11	Formal, plain-language	Unstructured RAG corpus - cited on request per US-11; source for the mandatory Data-handling-Note citation (Prompt v2.0 failure behavior, test T10)
SRC-010	Student-Support Case Agent Scope and Boundary Statement	Governance / AI use	Team-authored, derived directly from Project Charter 1, 4, 7 and Week 1 AI Boundary Matrix	Formal, plain-language	Unstructured RAG corpus - canonical grounding source for all boundary/redirect responses (US-5); highest-priority retrieval match for out-of-scope queries

SRC-011	BSSE Student Handbook and Departmental Guidelines	Academic curriculum	Synthetic, team-authored, modeled on Department of Computer Science (CoCIS) programme documentation	Semi-formal, institutional/technical	Unstructured RAG corpus - grounds course code, prerequisite, and support-pathway Q&A
SRC-012	Class Representative and Departmental Contacts (Redirect Directory)	Administrative - contacts	Synthetic, team-authored	Plain, directory-style	Unstructured RAG corpus - supplies the redirect_office value grounding referenced in Prompt v2.0 output schema
SRC-013	data/students.json	Deterministic student records	Synthetic, team-generated; no real student data (Project Charter 6-II)	N/A - structured data, not natural language	Deterministic JSON data source - read-only by the model; identity/ID validation is enforced in code, never by the LLM
SRC-014	data/cases.json	Deterministic case/ticket records	Synthetic, team-generated	N/A - structured data	Deterministic JSON data source - read/append only; no field in this file may be set to a status beyond “Pending” by the AI agent
SRC-015	data/timetable.json	Deterministic timetable records	Synthetic, team-generated	N/A — structured data	Deterministic JSON data source - read-only lookup grounding US-2; model must never state a timetable value not present here

Register size: 15 items (12 unstructured policy/governance documents, 3 deterministic JSON data sources) within the Project Charter's 10–50 document corpus target for Week 3.

Separation-of-concerns note: Doc IDs SRC-* and REG-* are retrieval-indexed, unstructured natural-language text intended for embedding/chunking and top-k retrieval. Doc IDs DATA-* are structured JSON, are never chunked or embedded into the vector index, and are accessed only through deterministic lookup functions (per the “LLM suggests, deterministic software enforces” rule).

2. Structured Corpus Excerpts

2.1 Unstructured policy documents (RAG corpus)

[DOC_ID: SRC-001]

[TITLE: Undergraduate Retake and Resit Policy]

[REGISTER: Formal, regulatory]

[GROUNDING_STATUS: Approved for grounded Q&A]

1. A student who fails a course unit may register to retake it in the next semester in which the course unit is offered, subject to departmental timetable capacity.
2. A retake attempt replaces the previous failing grade in the cumulative GPA calculation; the original attempt remains on the academic transcript.
3. A student may not retake a course unit more than twice without written permission from the Dean of the relevant college.
4. Retake registration follows the same add/drop deadlines published for the ordinary registration period (see SRC-003).
5. Regardless of underlying performance, the maximum grade recorded for a retaken course is capped at a Pass grade; the academic transcript explicitly marks the course as "retaken."

[DOC_ID: SRC-002]

[TITLE: Missing Marks and Incomplete Grade Procedure]

[REGISTER: Formal, procedural]

[GROUNDING_STATUS: Approved for grounded Q&A]

1. A "missing marks" case arises when a student's coursework or examination mark for a registered course unit does not appear on the provisional results list.
2. Students should first confirm with the course lecturer or departmental records office that the assessment was submitted and received.
3. If the discrepancy is not resolved informally, the student may submit a missing-marks case through the student-support channel, providing student ID, course code, semester, and a description of the missing assessment.
4. Missing-marks cases are reviewed and resolved by departmental academic staff; the support system records and routes the case but does not determine the outcome.
5. A missing-marks case must be submitted within four (4) weeks of the official publication date of provisional results; claims submitted after this window may be declined by departmental staff.
6. Every missing-marks case must include supporting evidence of coursework submission and/or examination attendance. The agent should prompt the student for this evidence when drafting a case but does not verify it.

[DOC_ID: SRC-004]

[TITLE: Student Support Case Categories and Escalation Routes]

[REGISTER: Formal, administrative]

[GROUNDING_STATUS: Approved for grounded Q&A and category suggestion]

1. Support cases submitted through the portal are classified into one of four standard categories: (a) Missing Marks, (b) Retake Request, (c) Registration Issue, or (d) Timetable Clash.
2. The agent may suggest the category that best matches a student's free-text description; the student confirms or corrects the suggestion before the case is finalised for staff review.
3. A case that appears disciplinary, admissions, or fee-related does not fit these four categories and must instead be routed as a boundary/redirect response per POL-010 and the contacts directory (REG-001), not created as a standard case.

[DOC_ID: SRC-009]

[TITLE: Data Handling and Student Privacy Note]

[REGISTER: Formal, plain-language]

[GROUNDING_STATUS: Approved cited on request per US-11]

1. Student data entered into the support assistant is used only to answer the student's question or process their support case.
2. No student data is sent to any service outside the approved application boundary (see Project Charter, Constraint III).
3. Students may ask, at any time, how their data is used; the assistant must answer using this note rather than a generated description.

[DOC_ID: SRC-010]

[TITLE: Student-Support Case Agent Scope and Boundary Statement]

[REGISTER: Formal, plain-language]

[GROUNDING_STATUS: Canonical boundary source - highest retrieval priority for out-of-scope queries]

1. This assistant answers questions about academic regulations, displays timetable and case-status information from approved records, and helps students describe and submit a new support case.
2. This assistant does not decide admissions, grading, disciplinary, or fee outcomes, and cannot change a case's status beyond "Pending."
3. Any request touching admissions, grading appeals outcomes, disciplinary decisions, or fee waivers is redirected to the relevant office listed in the departmental contacts directory (REG-001).
4. All case creation requires a name, student ID, category, and description; a case is only finalized for staff review after these fields are collected and confirmed.

[DOC_ID: SRC-011]

[TITLE: BSSE Student Handbook and Departmental Guidelines]

[REGISTER: Semi-formal, institutional/technical] [GROUNDING_STATUS: Approved for grounded Q&A]

1. The handbook lists course codes, prerequisite chains, and credit-unit weighting for the BSSE programme.
2. It records the standard support pathways a student should use for curriculum or course-content questions before escalating to a formal support case.

2.2 Deterministic synthetic data (never embedded or retrieved by the RAG pipeline)

// data/students.json - DATA-001, synthetic only, no real student records

```
[
  {
    "student_id": "2300700445",
    "name": "Isio Lily",
    "programme": "BSSE",
    "year_of_study": 4,
```

```

    "registration_status": "Registered",
    "academic_status": "Normal Progress"
  },
  {
    "student_id": "2300701098",
    "name": "Namatovu Christine Maria",
    "programme": "BSSE",
    "year_of_study": 4,
    "registration_status": "Registered",
    "academic_status": "Normal Progress"
  }
]

```

```

// data/cases.json - DATA-002, synthetic only
// AI BOUNDARY: "status" may only be written as "Pending" by any AI-facing
// write path. Any value beyond "Pending" requires a deterministic,
// human-approved action outside the model's reach.

```

```

[
  {
    "case_id": "CASE-0001",
    "student_id": "2300700445",
    "category": "missing_marks",
    "subject": "Missing Final Exam Score for BSE4104",
    "description": "Student reported that their final exam score for BSE4104 is missing on the portal.",
    "status": "Pending",
    "assigned_office": "Department of Computer Science",
    "created_by": "agent_draft",
    "created_at": "2026-09-05T09:14:00Z",
    "updated_at": "2026-09-05T09:14:00Z"
  },
  {
    "case_id": "CASE-0002",
    "student_id": "2300701098",
    "category": "registration_issue",
    "subject": "Add/Drop Deadline Conflict",
    "description": "Add/drop deadline conflict for elective course unit.",
    "status": "Pending",
    "assigned_office": "Department of Computer Science",
    "created_by": "agent_draft",
    "created_at": "2026-09-08T13:02:00Z",
    "updated_at": "2026-09-08T13:02:00Z"
  }
]

```

```

// data/timetable.json - DATA-003, synthetic only
[

```

```

  {
    "student_id": "2300700445",
    "semester": "2026-S1",
    "entries": [

```

```
{
  "course_code": "BSE4104",
  "day": "Tuesday",
  "time": "14:00-17:00",
  "room": "CoCIS LT1"
}
]
```

Provenance summary for auditability: Every SRC-* and REG-* item above is team-authored/synthetic text consistent with Project Charter 6 (assumptions) and 7 (constraints) - no restricted or copyrighted Makerere handbook text is reproduced or claimed as an official citation. All three DATA-* sources are synthetic and contain no real student, case, or timetable records, per Project Charter 6-II. This register should be committed to the repository alongside the corpus so retrieval provenance stays traceable to a specific Doc ID in every trace log (Week 1 US-9), and it remains fully compatible with the existing Week 3 RAG Evaluation Set (Q01–Q15) without any Doc ID or case ID changes.